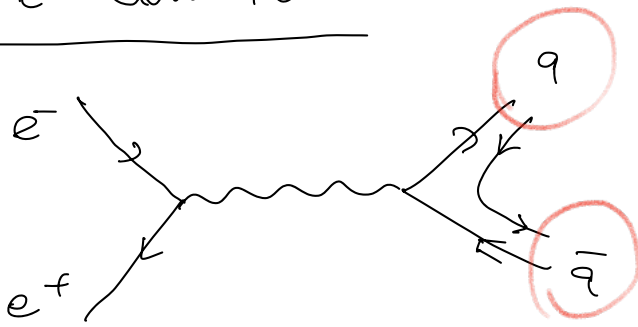
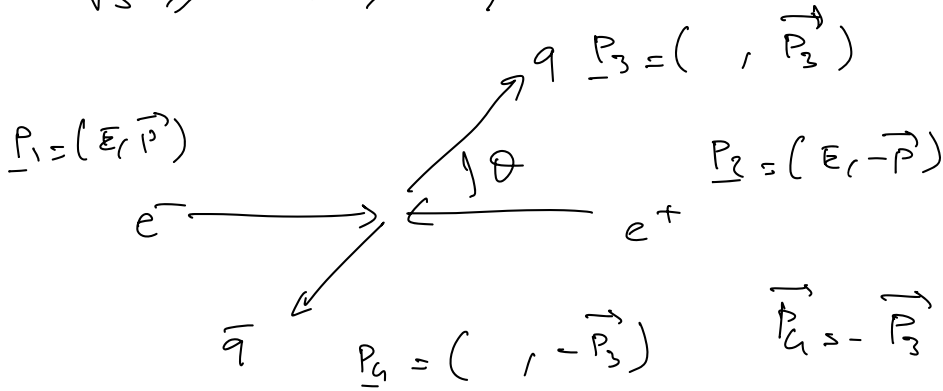


e^+e^- collisions

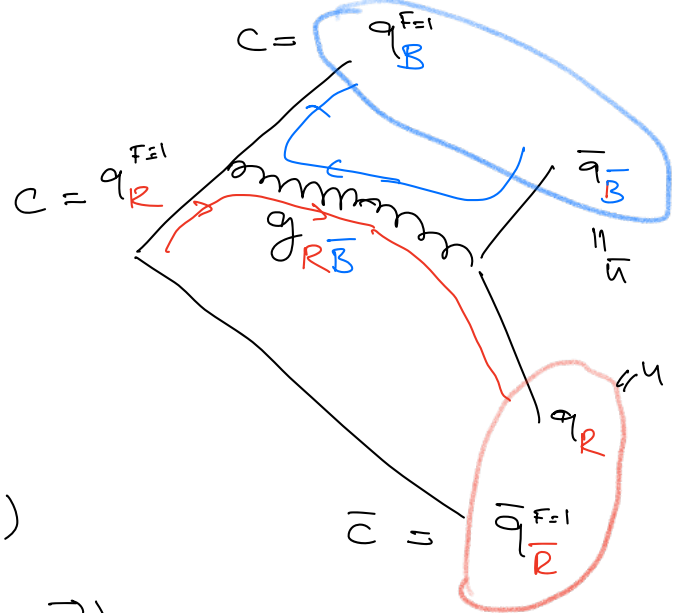


$$\sqrt{s} \gg 2m_q \gg 2m_\mu = 200 \text{ MeV}$$



$$E \gg m_e, m_q$$

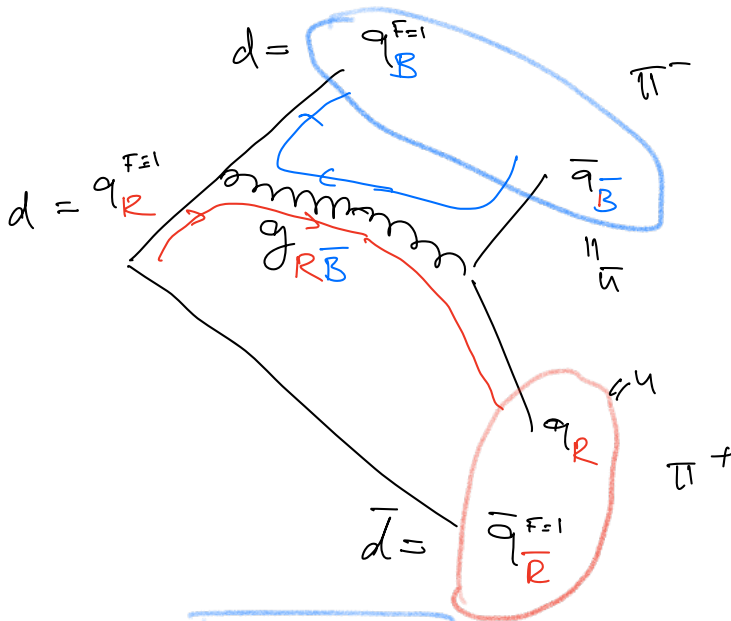
$$|\vec{P}_3| = |\vec{P}_4| \approx E$$



$$e^+e^- \rightarrow D^+ D^-$$

$$m_D \approx 1.864 \text{ GeV}$$

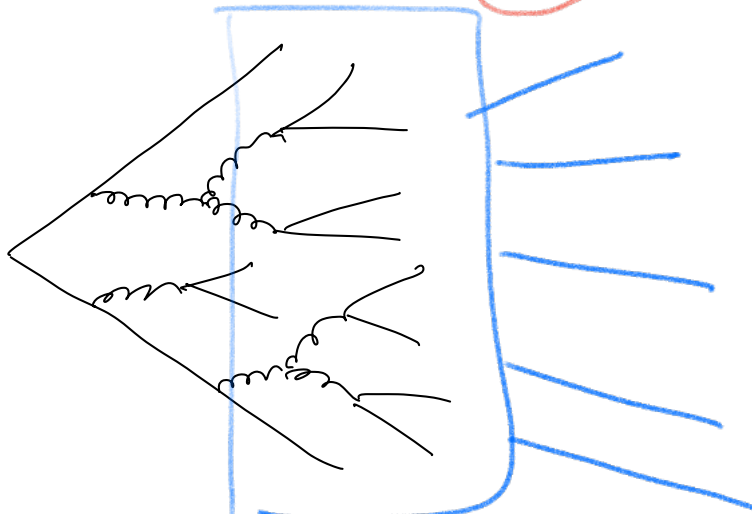
$$\sqrt{s} \gg 2 \times 1.864 \text{ GeV} = 3.7 \text{ GeV}$$



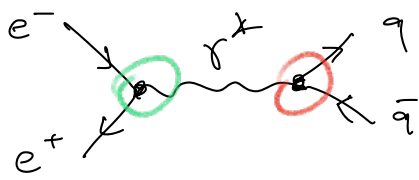
$$e^+e^- \rightarrow \pi^+ \pi^-$$

$$\sqrt{s} \gg 280 \text{ MeV}$$

$$e^+e^- \rightarrow \text{hadrons}$$

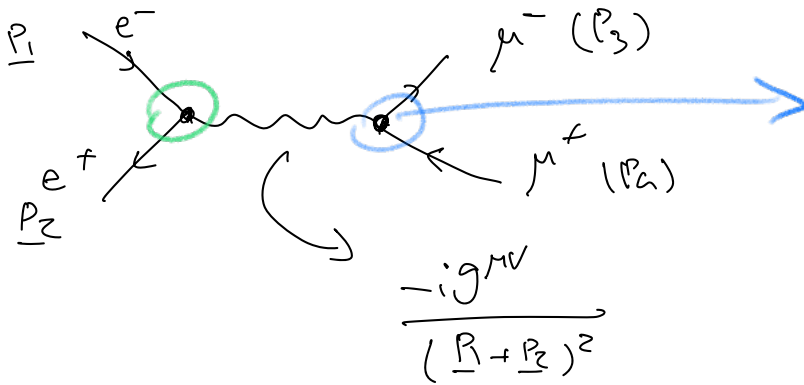


Hadronization



$$\sigma(e^+e^- \rightarrow q\bar{q}) = ?$$

ieZ_q Z_q : electric charge of quark q



$$\bar{u}(p_3) ie \gamma^\nu v(p_4)$$

$$\sqrt{s} \gg 2m_\mu.$$

$$E_1 = E_2 = E \gg m_\mu, m_e, m_q$$

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2} \frac{1}{(E_1 + E_2)^2} \frac{|\vec{p}_{out}|}{|\vec{p}_{in}|} |M|^2$$

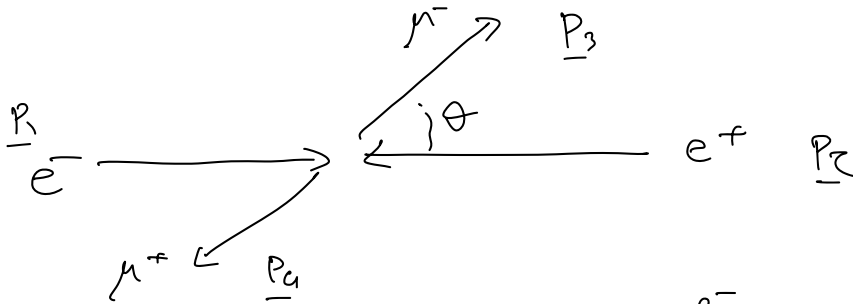
$$E_3 = E_4 = E$$

$$\Rightarrow \frac{|\vec{p}_{out}|}{|\vec{p}_{in}|} = 1$$

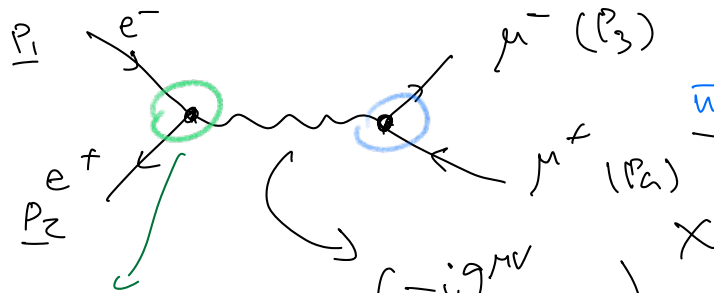
$$(E_1 + E_2)^2 =: s = (p_1 + p_2)^2$$

$$p_1 = (E_1, \vec{p})$$

$$p_2 = (E_1, -\vec{p})$$



$$|M|^2 = ?$$



$$\bar{u}(p_3) ie \gamma_\mu v(p_4)$$

$$-iM = \left(\bar{v}(p_2) \gamma_\mu u(p_1) \right) \times \left(\frac{-ig^{\mu\nu}}{(p_1 + p_2)^2} \right) \times$$

$$e = \sqrt{4\pi\alpha}$$

$$e^2 = 4\pi\alpha$$

$\langle |M|^2 \rangle =$ average over initial state spins, sum over final state spins.

Initial state spin configurations

$$e_L^- \xrightarrow{\leftarrow} p \xleftarrow{S \leftarrow} e_R^+$$

$$S_z = -1 \quad (\text{total spin})$$

$$S \leftarrow \rightarrow p(\text{momentum})$$

$$h = \frac{\vec{S} \cdot \vec{P}}{|\vec{S}| |\vec{P}|}$$

$$e_R^- \Rightarrow \leftarrow \Rightarrow e_L^+$$

$$S_z = +1$$

$$e_R^- \Rightarrow \leftarrow \leftarrow e_R^+$$

$$S_z = 0$$

$$e_L^- \xrightarrow{\leftarrow} \leftarrow \Rightarrow e_L^+$$

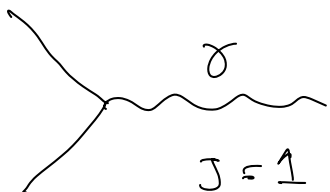
$$S_z = 0$$

not allowed.

4 initial configurations.

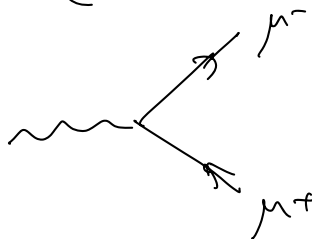
we are in massless regime ($E \gg m_e, m_\mu$)

$$e_L^- e_R^+$$



$$J = 1$$

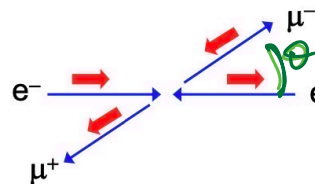
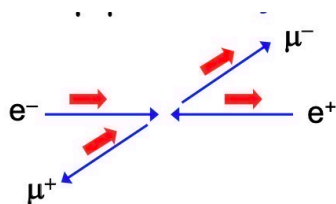
$$J_z = \pm 1$$



only consider initial states
 $S_z = \pm 1$

Final states.
with $S_z = \pm 1$

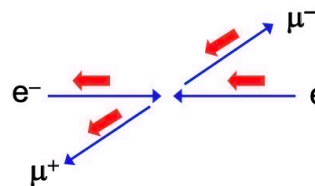
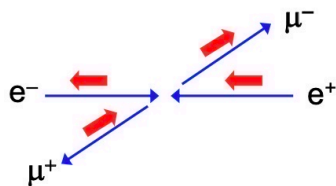
$$M_{RR}$$



$$M_{RL}$$

$$\propto \frac{1 - \cos \theta}{2}$$

$$M_{LR}$$



$$M_{LL}$$

$$\propto \frac{1 + \cos \theta}{2}$$

$M_{\alpha\beta}$:
 α : helicity of e^-
 β : helicity of μ^-

$$R = \Rightarrow$$

$$L = \leftarrow$$

look at $\theta = 0$

M_{RR}

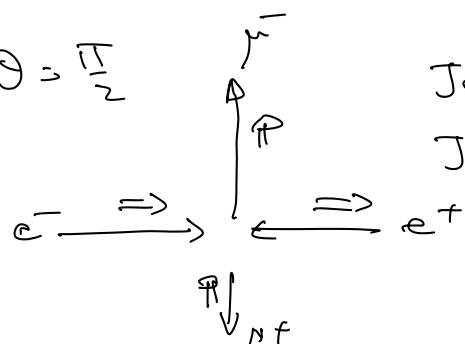
$$J_z^{\text{in}} = +1$$

$$e^- \Rightarrow \leftarrow \Rightarrow e^+$$

$$J_z^{\text{fin}} = +1$$

$$\mu^+ \leftarrow \Rightarrow \Rightarrow \mu^-$$

$$\theta = \frac{\pi}{2}$$



$$J_z^{\text{in}} = +1$$

$$J_z^{\text{fin}} = 0$$

$$\begin{array}{ccc} e^- & \xrightarrow{=} & e^+ \\ \mu^- & \xleftarrow{=} & \mu^+ \end{array}$$

$$\theta = \pi$$

$$\begin{array}{l} J_z^m = +1 \\ J_z^{\mu} = -1 \end{array}$$

$$M_{LL} = M_{RR} = (4\pi\alpha) \frac{1 + \cos\theta}{2}$$

$$\text{for } \theta \rightarrow \pi \Rightarrow M_{RR} \Rightarrow 0.$$

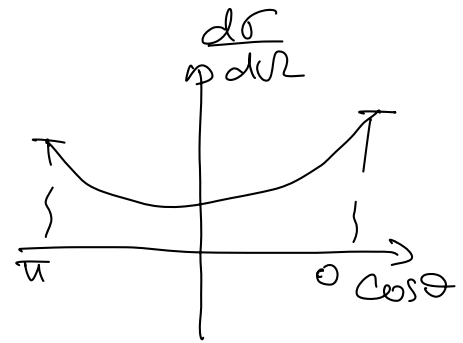
$$M_{RL} = M_{LR} = 4\pi\alpha \frac{1 - \cos\theta}{2}$$

$$\langle |M|^2 \rangle = \frac{1}{4} \left(|M_{LL}|^2 + |M_{RR}|^2 + |M_{LR}|^2 + |M_{RL}|^2 \right) (4\pi\alpha)^2$$

initial state configurations

$$\left(\begin{array}{l} 2(1 + \cos\theta)^2 + 2(1 - \cos\theta)^2 \\ 2(1 + 2\cancel{\cos\theta} + \cos^2\theta) + \\ 2(1 - 2\cancel{\cos\theta} + \cos^2\theta) \end{array} \right)$$

$$\begin{aligned} \langle |M|^2 \rangle &= \frac{1}{4} (4\pi\alpha)^2 4(1 + \cos^2\theta) \\ &= (4\pi\alpha)^2 (1 + \cos^2\theta) \end{aligned}$$



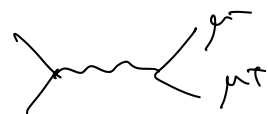
$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2} \frac{1}{(E_1 + E_2)^2} (4\pi\alpha)^2 (1 + \cos^2\theta)$$

$$d\Omega = 2\pi d\cos\theta.$$

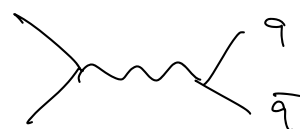
$$\sigma_{\text{tot}} = \int \left(\frac{d\sigma}{d\Omega} \right) d\Omega = \frac{16\pi}{3} \frac{\alpha^2}{s} \quad e^+e^- \rightarrow \mu^+\mu^-$$

$$s = (E_1 + E_2)^2 = (2E)^2 = 4E^2 \quad E: \text{energy of the beam.}$$

$$\sigma(e^+e^- \rightarrow \mu^+\mu^-) = \frac{16\pi}{3} \frac{\alpha^2}{s}$$



$$\sigma(e^+e^- \rightarrow q\bar{q}) = \frac{16\pi}{3} \frac{\alpha^2}{s} Q_q^2$$



For one flavor

Experimentally measure

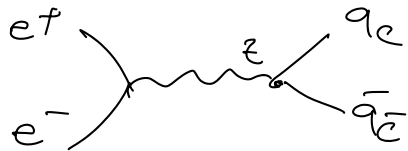
$$R = \frac{\sigma(e^+e^- \rightarrow \text{hadrons})}{\sigma(e^+e^- \rightarrow \mu^+\mu^-)} = \frac{\# \text{ events w/ had}}{\# \text{ events w/ } \mu^+\mu^-}$$

$$R = \sum_i^{\text{flavors}} Z_{q_i}^2$$

Sum over all flavors q_i with $\sqrt{s} \geq 2m_{q_i}$

Number of flavors depends on $\sqrt{s}$

all quarks with $\sqrt{s} \geq 2m_{q_i}$ are accessible.



$$\mu^2 \sim Z_q^2 \text{ for a given color.}$$

$$|M_{q\bar{q}}|^2 = |M_{\bar{u}u}^B|^2 + |M_{\bar{u}u}^G|^2 + |M_{\bar{u}u}^R|^2$$

$$\Rightarrow \sigma(e^+e^- \rightarrow q\bar{q}) = N_c \sigma(e^+e^- \rightarrow q_c \bar{q}_c) \\ = \underbrace{N_c}_{\# \text{ colors.}} \underbrace{(\sum_i Z_{q_i}^2)}_{\text{Flavors}} \underbrace{\sigma(e^+e^- \rightarrow \mu^+\mu^-)}_{\text{QED.}}$$

$$R = \frac{\sigma(e^+e^- \rightarrow \text{had})}{\sigma(e^+e^- \rightarrow \mu^+\mu^-)} = N_c \times \sum_{\sqrt{s} \geq 2m_{q_i}} Z_{q_i}^2$$

$$\sqrt{s} \geq 2m_u, 2m_d, 2m_s.$$

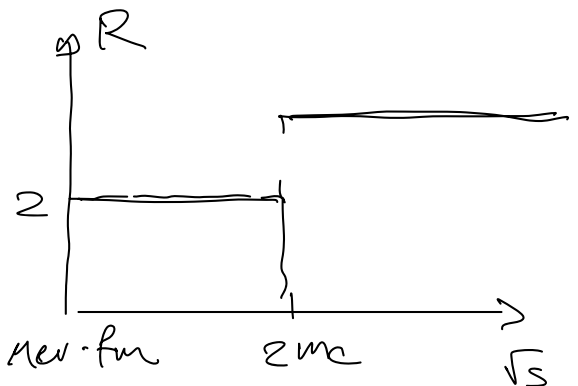
suppose $N_c = 3$

$$\begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} C \\ S \end{pmatrix}^{Q + \frac{2}{3}}$$

$$R = 3 \times \left(\frac{4}{9} + \frac{1}{9} + \frac{1}{9} \right) = 3 \cdot \frac{6}{9} = 2 \quad \text{until } \sqrt{s} \leq 2m_c$$

$$\text{if } \sqrt{s} \geq 2m_c$$

$$R = 2 + 3 \times \left(\frac{2}{3} \right)^2 = 2 + \frac{12}{9}$$



$$\sigma = \frac{16\pi}{3} \frac{\alpha^2}{s}$$

$$\alpha = \frac{1}{137}$$

$$\hbar c = 197 \text{ MeV} \cdot \text{fm}$$

$$\sigma = \frac{86.8 \text{ nb}}{S [\text{GeV}^2]}$$

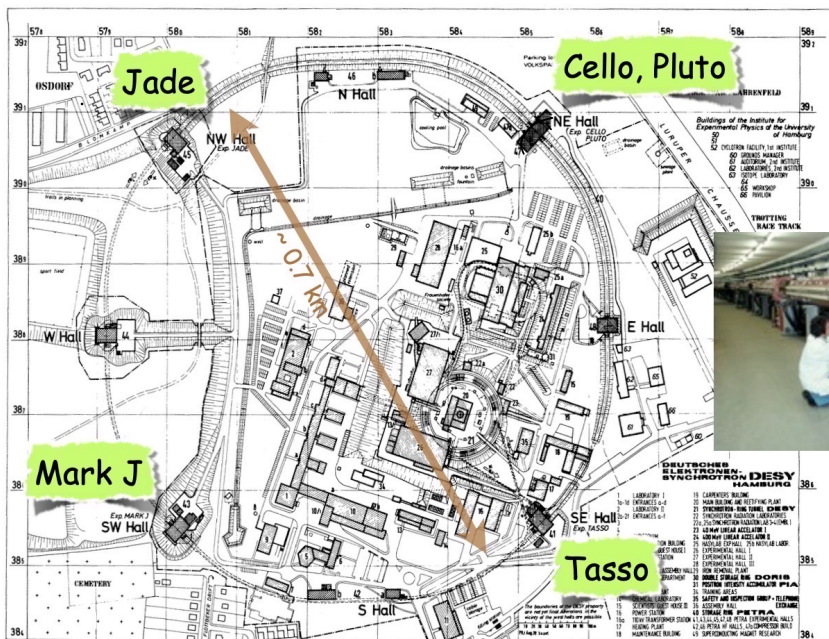
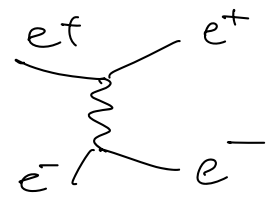
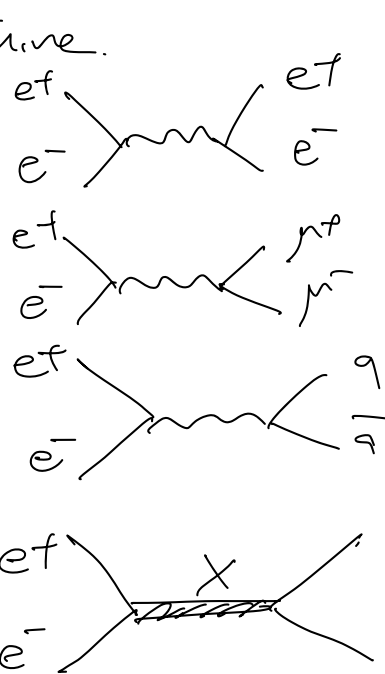
$$= \frac{(86.8)}{4} \frac{1}{E_{\text{beam}}^2 [\text{GeV}]}$$

nb.

$$S = 4 E_{\text{beam}}^2$$

$\sqrt{s}$ fixed by the machine.

$e^+e^- \rightarrow e^+e^-$
 $\mu^+\mu^-$
 hadrons.
 X



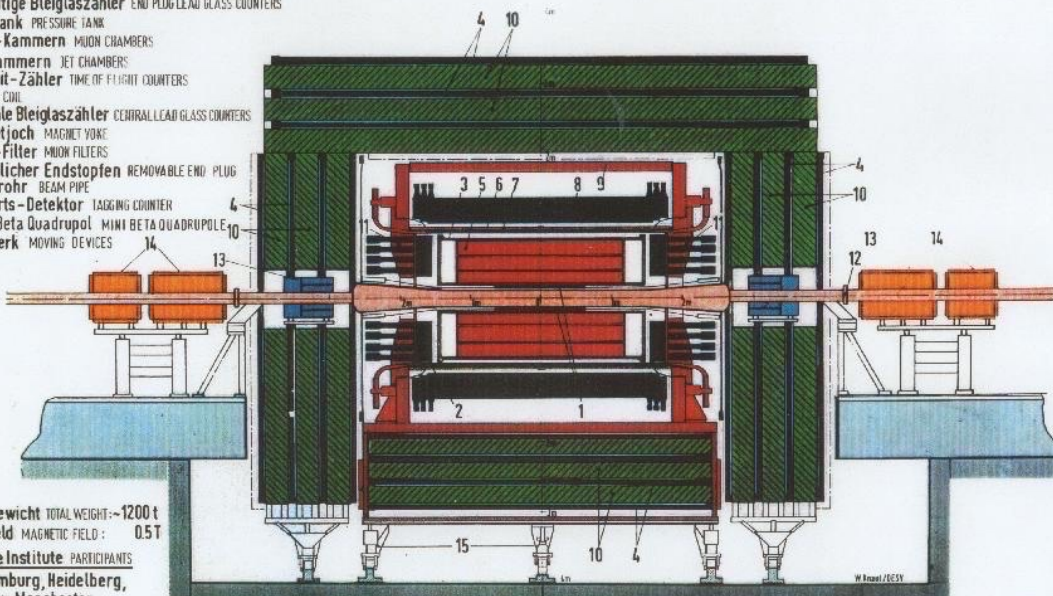
PETR @ DESY
 near Hamburg

1976 → 1986

$\sqrt{s}: 12 \rightarrow 46.7 \text{ GeV}$

MAGNETDETEKTOR JADE

- 1 Strahlrohrzähler BEAM PIPE COUNTERS
- 2 Endseitige Bleiglaszähler END PLUG LEAD GLASS COUNTERS
- 3 Drucktank PRESSURE TANK
- 4 Myon-Kammern MUON CHAMBERS
- 5 Jet-Kammern JET CHAMBERS
- 6 Flugzeit-Zähler TIME OF FLIGHT COUNTERS
- 7 Spule COIL
- 8 Zentrale Bleiglaszähler CENTRAL LEAD GLASS COUNTERS
- 9 Magnetjoch MAGNET YOKE
- 10 Myon-Filter MUON FILTERS
- 11 Beweglicher Endstopfen REMOVABLE END PLUG
- 12 Strahlrohr BEAM PIPE
- 13 Vorwärts-Detektor FORWARD COUNTER
- 14 Mini-Beta Quadrupol MINI BETA QUADRUPOLE
- 15 Fahrwerk MOVING DEVICES



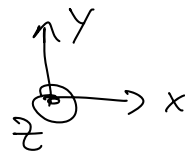
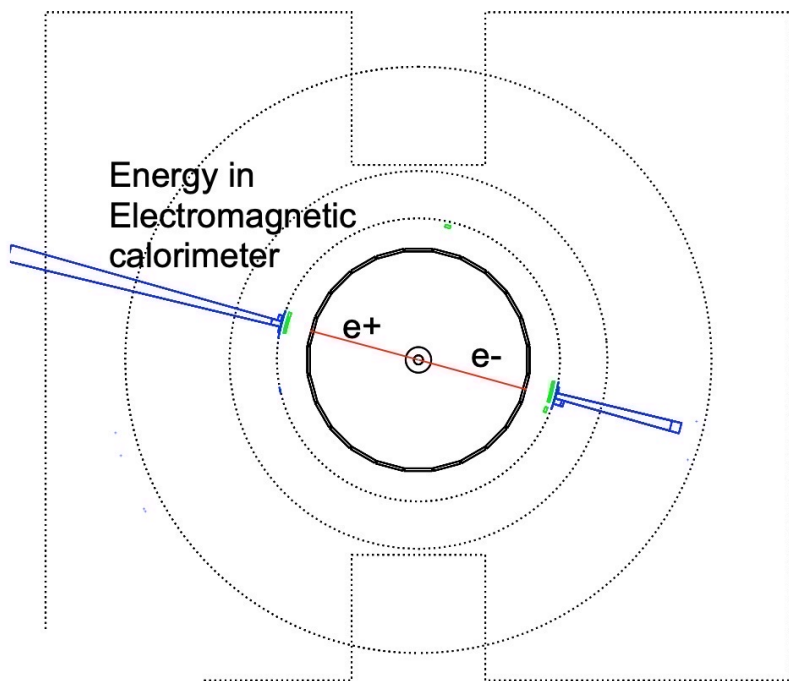
Gesamtgewicht TOTAL WEIGHT: ~1200 t
 Magnetfeld MAGNETIC FIELD: 0.5 T

Beteiligte Institute PARTICIPANTS
 DESY, Hamburg, Heidelberg,
 Lancaster, Manchester,
 Rutherford Lab., Tokio

Japan
 Deut che land
 Eng land
 Defector

How to distinguish

e^+e^- , $\mu^+\mu^-$, hadrons.



Initial state e^+e^-

$$\vec{P}_T = \vec{0}$$

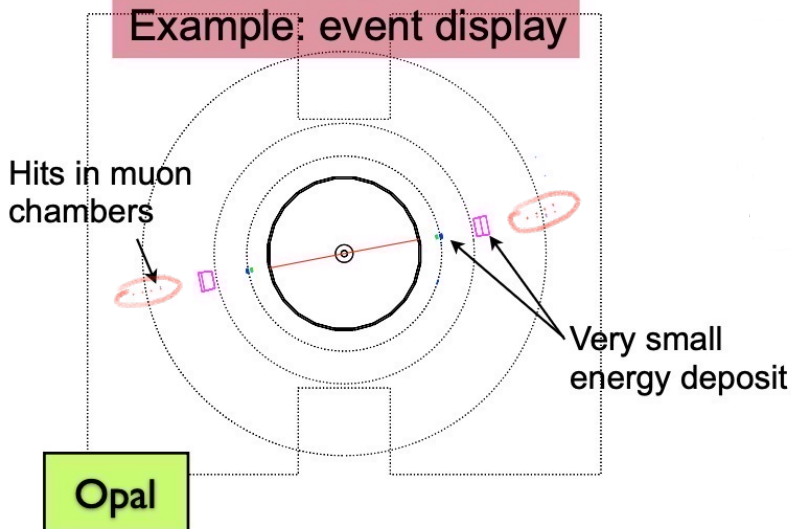
$\Rightarrow$ Final state

$$\sum_i \vec{P}_{Ti} = \vec{0}$$

Back-to-back charged tracks.

+ Energy deposit in EM Calorimeter.

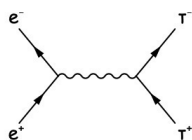
Example: event display



back-to-back tracks.
no or little deposit in EM Calorimeter.

hits in muon chambers.

$$e^+e^- \rightarrow \mu^+\mu^-$$



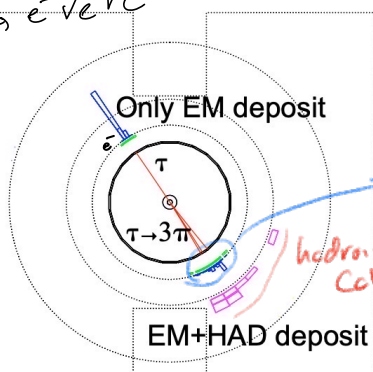
$$\tau \rightarrow e \bar{\nu}_e \nu_\tau$$

$$M_\tau = 1.76 \text{ GeV}$$

hadronic decays of τ lepton.

$$\tau \rightarrow 3\pi$$

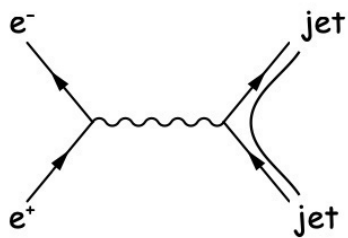
$$\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau$$



EM Calorimeter

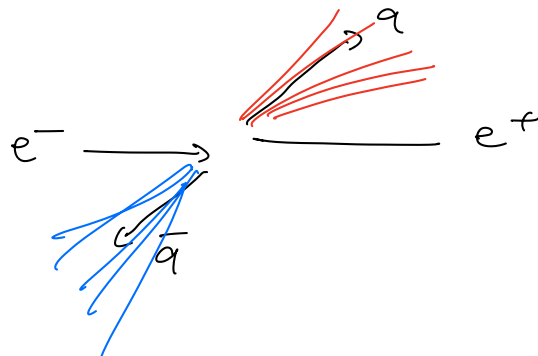
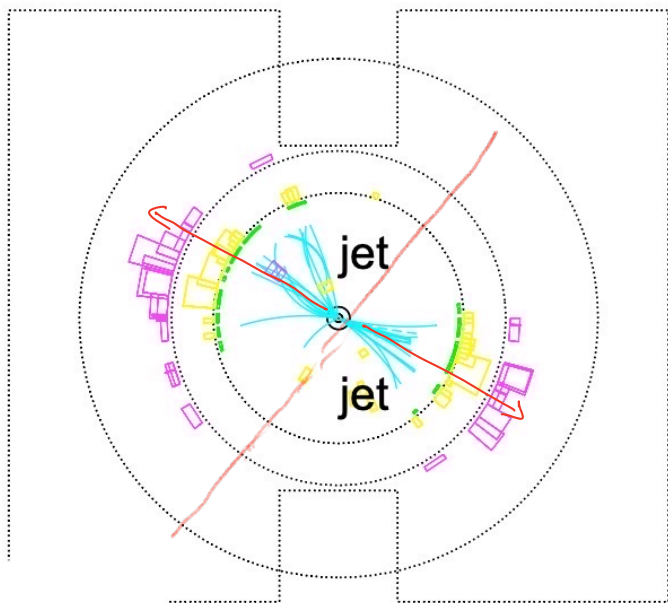
hadronic Calorimeter.

EM+HAD deposit

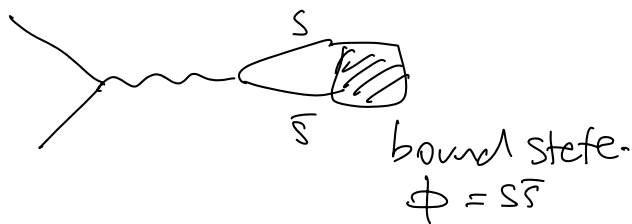


$e^+e^- \rightarrow \text{hadrons}$

Many charged tracks.
Many deposits

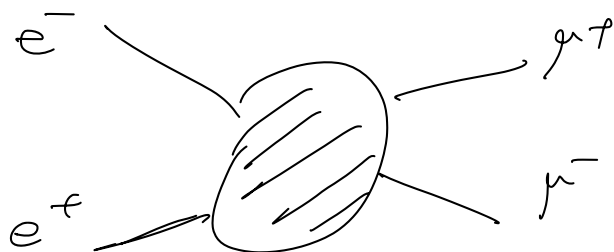


$e^+e^- \rightarrow s\bar{s}, c\bar{c}$



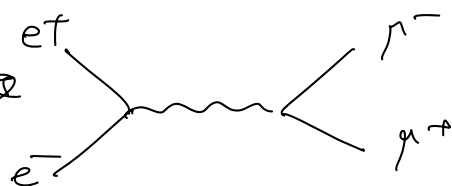
$c\bar{c} = J/\psi$

$\sigma(e^+e^- \rightarrow X \rightarrow q\bar{q}, l\bar{l})$



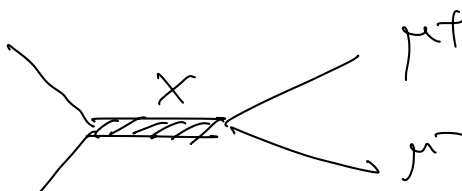
Experimental view.

just QED:



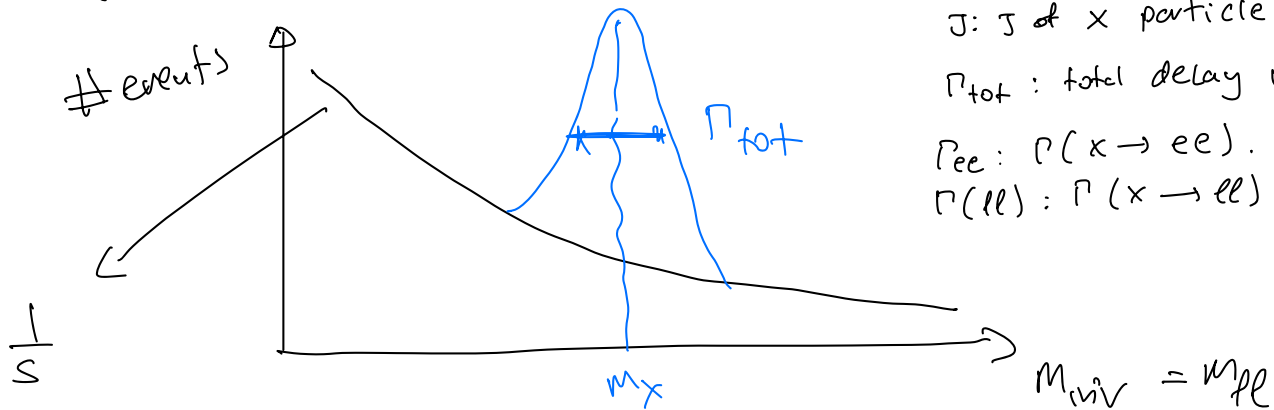
$$\sigma \sim \frac{16\pi}{3} \frac{\alpha^2}{s}$$

or production of X



$$\sigma(e^+e^- \rightarrow X \rightarrow e^+e^-) = \frac{16\pi}{s} \frac{2J+1}{(2S_1+1)(2S_2+1)} \frac{\Gamma_{ee}}{\Gamma_{\text{tot}}} \frac{\Gamma_{ee}}{\Gamma_{\text{tot}}} \frac{\Gamma_{\text{tot}}^2}{(m_X - \sqrt{s})^2 + \Gamma_{\text{tot}}^2/4}$$

$$M_{ee} = \sqrt{(p_{e^+} + p_{e^-})^2} \equiv m_X \text{ if from decay of } X$$



if $\sqrt{s}$ near $m_X \pm \Gamma \Rightarrow$ see resonance

$$N_{\text{events}} = \underbrace{\sigma}_{\text{depends on the physics process}} \cdot \underbrace{X}_{\text{inst. lum: property of the collider}} \cdot \underbrace{\Delta t}_{\text{data taking time}}$$

$$N_{e^+e^-} = \sigma_{e^+e^-} (L_{\text{inst}} \Delta t)$$

$$N_{\text{had}} = \sigma_{\text{had}} (L_{\text{inst}} \Delta t)$$

$$N_{\mu\mu} = \sigma_{\mu\mu} (L_{\text{inst}} \Delta t)$$

in common for all processes.

$$N_{\text{had}} = \sigma_{\text{had}} \cdot \frac{N_{e^+e^-}}{\sigma_{e^+e^-}}$$

$N_{e^+e^-}$ → count # events.
 $\sigma_{e^+e^-}$ → compute from theory.
 Bhabha scattering

measurement of luminosity w/ Bhabha events

$$\sigma_{\text{had}} = \underbrace{\frac{N_{\text{had}}}{N_{e^+e^-}}}_{\text{Ratio of exp. quantities}} \sigma_{e^+e^-}$$